Roman Losan

Senior AI Enablement Engineer

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Date of birth: 17.02.1990 | Citizenship: Latvia

Senior engineer fluent in English and Russian, focused on AI-assisted development enablement: coaching engineers, building internal AI tooling and MCP integrations, and measuring real productivity gains. Strongest work: agent harness systems (multi-agent orchestration, LLM-judge rubrics, self-evolution, +74% pass rate), a unified LLM gateway serving 22+ models with RAG and persistent long-term memory, and low-latency Rust systems (Solana bot, 82ms). Strong in Python, Node/TypeScript, MCP, RAG, prompt engineering, and AI tooling (Claude Code, Cursor, Codex).

Experience

Girasol — AI Agent Engineer

2024 — Present

- Designed and deployed a production multi-agent system with 6+ autonomous agents — orchestrator + specialists (backend, frontend, research, devops) — each with its own LLM model, system prompt, and skill set. Model routing by task complexity: cheap models for routine, strong models for reasoning — saving tokens without quality loss.
- Built a Solana sniper/copytrade bot in Rust with **82ms end-to-end latency** — ShredStream (189ms), gRPC Geyser (0.94ms), Jito bundles, PumpFun/Raydium integration, position manager (take-profit/stop-loss), covered by tests.
- Set up autonomous coding pipeline (OpenCode/MiMo + GSD methodology): decompose → execute → verify → ship. TDD mandatory, code without passing tests not accepted. Multi-step Rust milestones through systemd services without manual intervention.
- Engineered a unified LLM API proxy (CLIProxyAPI) serving **22+ models** from multiple providers (OpenAI, Anthropic, Google, Ollama) through a single OpenAI-compatible endpoint with credential rotation — **$0 API cost**.
- Deployed and managed local LLM models for diverse tasks — text generation (GLM, DeepSeek, Kimi, Qwen), embeddings (granite-embedding:278m), TTS (edge-tts), vision (Gemini proxy), code generation (Kimi K2.7 Code) — eliminating SaaS dependencies.
- Implemented agent self-evolution with weekly cycles — OPRO prompt optimization, LLM-judge rubric (5-axis), Agent Evaluation — improving constraint pass rate from **0.575 to 1.000 (+74%)**.
- Diagnosed non-trivial LLM bug: reasoning models via streaming put all output in `reasoning` field, leaving `content` empty at low token budgets. Localized via SSE-delta analysis, separated gateway issue from response format issue.
- Built persistent agent memory: SQLite + FTS5, RAG hybrid search (LightRAG, 6 retrieval modes), knowledge graph, auto-context injection on session start and after compaction. Anti-amnesia architecture — facts surviving context compression.
- Built web dashboard in TypeScript/React — wallet tracker, Kanban board, end-to-end backend integration.
- Integrated MCP servers (12 broker tools, CodeGraph AST knowledge graph, MiMo Code agent). DevOps: systemd, Docker, cron monitoring, WSL2 stabilization (OOM/cgroup memory leaks), watchdog auto-restart.

SoftBlues — AI Pipeline Engineer

2023 — 2024

- Designed and iterated an AI-assisted image processing pipeline for high-volume visual content workflows — **hundreds of images per day** with automated visual decisions.
- Developed rule-based visual logic on OpenCV/DIMA grid detection: occupied columns, edge selection, centering people, avoiding body-part cuts, choosing crop boundaries by visual rules.
- Integrated and evaluated CV/AI components: GAIC, CLIP/few-shot scoring, Florence-2, OpenCV, ONNX/OpenVINO, SAM2-style segmentation.
- Designed pipeline principle: analyze reduced-size images for speed, then apply final crop coordinates to original-resolution images.
- Human-in-the-loop feedback: compared results against user-approved examples, converted corrections into reusable rules. Golden-reference testing — new logic must not break previously approved outputs.
- Debugged long-running pipelines: subprocess timeouts, GAIC hangs, disk I/O stalls, partial outputs, safe retry of only failed items.

Languages

Russian (native)
English — B2 (Upper-Intermediate)
Spanish — B2 (Upper-Intermediate)

Technical Skills

Programming

Python, Rust

NLP & ML

NLP, Machine Learning, Large Language Models, PyTorch (evaluated), Transformers, fine-tuning, LLM post-training, inference optimization, generative AI, sentiment analysis, emotion detection, dialog management, multi-turn dialogue, response generation, retrieval-augmented generation

Computer Vision

OpenCV, CLIP/few-shot scoring, Florence-2, GAIC, SAM2, ONNX/OpenVINO (evaluated), image alignment, candidate generation, hard-gate validation, golden-reference testing

Agent Frameworks

LangGraph (State, Nodes, Edges, Checkpointing), LangChain (evaluated), Multi-agent Orchestration, CrewAI (evaluated), AutoGen (evaluated), LLM Orchestration, Function Calling, Human-in-the-Loop

LLM Platforms

OpenAI API (GPT-5.x, Function Calling, Tool Use), Anthropic (Claude 4.x), Google Gemini, Ollama (GLM, DeepSeek, Kimi, Qwen), Prompt Engineering, unified API proxy (22+ models, multi-provider), local LLM deployment (TTS, voice synthesis, vision, image processing, embeddings, code gen), multimodal agents

RAG & Vector Databases

LightRAG (6 retrieval modes), GBrain PGLite, Graphiti, Cognee, FalkorDB, Pinecone (evaluated), Weaviate (evaluated), Vector Database, Hybrid Search, BM25, Reranking, Ollama Embeddings

Memory & Personalization

Long-term memory, conversational memory, persistent SQLite+FTS5, knowledge graph, context

Freelance — Traffic Arbitrage & Digital Marketing

~2014 — 2024

- Managed digital advertising and traffic arbitrage campaigns across multiple platforms, optimizing acquisition costs and conversion metrics for private clients.
- A/B testing, audience targeting, landing page optimization. Full P&L responsibility.

IBM Latvia — Lotus Notes / Domino Administrator

~2012 — 2014

- Administered Lotus Notes/Domino server infrastructure — mail routing, application servers, user access management, backups, troubleshooting.

Education

Riga Technical University (RTU), Latvia — Faculty of Information Technology, 2012
.NET Development Course, Accenture, Latvia, 2013

Interests

Cryptocurrency trading and on-chain analytics
Applied AI research — autonomous agents, memory architectures
Culinary arts

injection, personalization, avatar behavior, anti-amnesia, trust tiers

MCP Integration

stdio transport, SSH cross-host MCP, broker pattern (12 tools), CodeGraph (AST knowledge graph), MiMo Code agent

Infrastructure & DevOps

Docker, Kubernetes (evaluated), systemd, FastAPI, Redis, SQLite, PGLite, WSL2 stabilization (OOM, cgroup), watchdog, cron, health checks, cost optimization

Observability & Evals

Custom L7 evaluation suite, metric passports, recall precision testing, LangSmith (evaluated), Agent Evaluation, OpenTelemetry (evaluated)